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ARTICLE

Modeling of the Helium-Heated Steam Reformer for HTR-10

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In this study, based on the pseudo-homogeneous one-dimensional model, a steady-state model of the helium-heated steam reformer planned to be connected with the 10 MW high temperature gas cooled reactor (HTR-10) has been developed. Good agreement is shown between the simulating results and experimental data. The influence of main process parameters on the performance with respect to the methane conversion and the hydrogen yield is investigated and discussed. The performance increases remarkably with the increase in the inlet helium temperature when it is lower than 1,000°C. Whereas, the effect becomes weak when the temperature is higher than 1,000°C. The influence of the inlet helium flow rate is not as evident as that of the temperature. The inlet helium pressure and inlet process gas temperature have almost no influence on the performance. The performance increases with the decrease in the inlet process gas pressure. The influence of the inlet process gas flow rate and steam-to-carbon ratio (S/C) is complicated. Optimal values should be chosen for them to obtain a high performance.

KEYWORDS: nuclear process heat application, HTR-10, hydrogen production, helium-heated steam reformer

I. Introduction

World hydrogen consumption for fertilizer and oil refining is about 50 million metric tons per year.[1] The rapid growth in demand is a result of the decreased availability of the light crude oil and corresponding increased use of the heavy crude oil.[2] The light crude oil does not require extra hydrogen for conversion to gasoline while the heavy crude oil requires a massive amount of hydrogen for conversion to gasoline. If the cost goals for automotive fuel cells are reached, the transportation sector may ultimately be fueled by hydrogen. This implies a growth in the hydrogen consumption by one to two orders of magnitude over a period of several decades.[3] Although the demand for hydrogen will increase in the near future, a primary energy is needed to produce hydrogen from the fossil fuel or water. Taking into account global warming and environment protection issues, nuclear energy supplied by the high-temperature gas-cooled reactor is a suitable energy source for a huge amount of hydrogen production compared with the fossil fuel energy.[4]

HTR-10 was operated in 2000 at the Institute of Nuclear and New Energy Technology (INET), Tsinghua University, Beijing. This test reactor will be utilized to develop a modular high temperature gas cooled reactor technology as well as to establish an experimental base for a nuclear process heat application. INET has started a R&D project for developing a hydrogen system.[5] A hydrogen production system by steam methane reforming (SMR) is to be the first heat utilization system of HTR-10. The coupling technology between HTR-10 and the hydrogen production system by SMR can contribute to other hydrogen production systems such as water-splitting by a thermochemical method.[6,7]

The helium-heated reformer is the key component of the hydrogen production system. It is different from that of the conventional reformer in such aspects as heat transferring, geometry and operation conditions. It was experimentally investigated in the EVA-I and EVA-II test facilities at the Research Center Juelich[8] and in the mock-up model test facilities at Japan Atomic Energy Agency (JAEA).[9] In this paper, a systematic analysis of the performance of the reformer by means of numerical simulations is addressed. A steady-state mathematical model based on the pseudo-homogeneous one-dimensional model has been set up and a corresponding computer program is developed. With this program, the influence of main process parameters such as the inlet helium parameters, inlet process gas parameter, and S/C on the performance with respect to the methane conversion and hydrogen yield is investigated and discussed.

II. HTR-10 Hydrogen Production System

1. Overview of HTR-10 Hydrogen Production System

The HTR-10 hydrogen production system is designed to utilize the nuclear heat effectively and to achieve hydrogen productivity competitive to that of a commercial fossil-fired plant with operability, controllability, and safety. **Figure 1**

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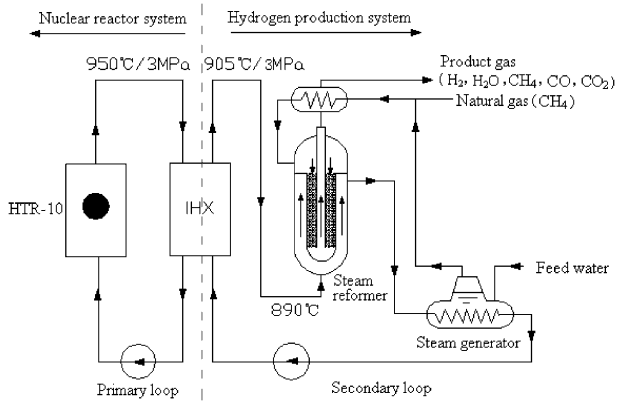


Fig. 1 Flow scheme of the HTR-10 hydrogen production system

shows an arrangement of main components of the HTR-10 hydrogen production system. HTR-10 can supply the nuclear heat with 750°C now, and later with 950°C for the hydrogen production, to intermediate heat exchanger (IHX) in a primary helium gas loop. And the nuclear heat is transferred from the primary helium gas loop to a secondary helium gas loop at IHX, which is to be utilized as a heat source for the hydrogen production. Hydrogen is produced by SMR in a steam reformer (SR) which is designed to have an approximate hydrogen production capacity of 2,500 Nm^3/h. Due to the heat loss along the secondary helium gas pipes from IHX to SR, the secondary helium gas temperature is reduced to 890°C at SR inlet whereas the IHX outlet temperature is 905°C.

2. Helium-heated Steam Reformer

The helium-heated steam reformer is the key component of the HTR-10 hydrogen production system. Two main chemical reactions are considered here.

The first reaction is the endothemic steam reforming reaction:

$$CH_4 + H_2O = CO + 3H_2 - 205.8\,kJ/mol. \quad (1)$$

The second reaction is the exothemic water gas shift reaction:

$$CO + H_2O = CO_2 + H_2 + 41.2\,kJ/mol. \quad (2)$$

The overall reaction is as follows:

$$CH_4 + 2H_2O = CO_2 + 4H_2 - 164.6\,kJ/mol. \quad (3)$$

Although the second reaction is exothemic, the overall reaction is endothemic. Hence, the conversion requires a large quantity of heat. And the number of the moles of the conversion is increased. So, high-temperature and low-pressure conditions are more favorable for this process. In addition, this process is limited by a thermodynamic equilibrium. The structure schematic diagram of the reformer is illustrated in **Figs. 2**(a) and (b). Generally speaking, the helium-heated reformer is a tube-shell type reformer which is made up of reformer tube bundles. From inside to outside, it can be divided into three parts: centric tube, catalyst tube, and guide tube. The process gas with a low temperature enters the process gas inlet at the top and flows downwards through

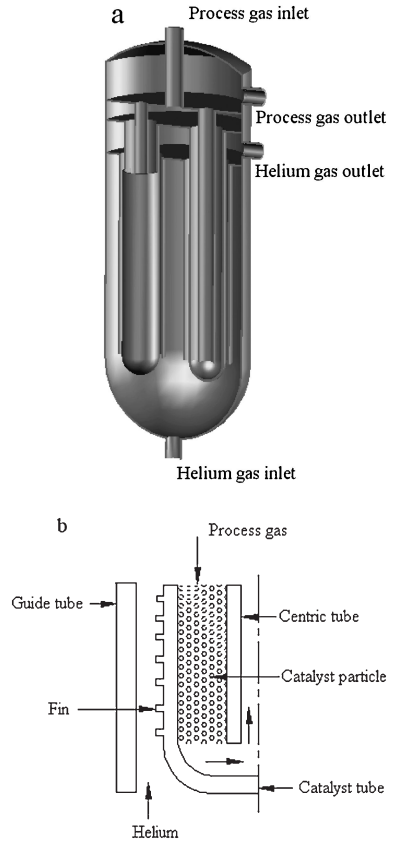


Fig. 2 Schematic diagram of the steam reformer

the annular catalyst bed, where it absorbs two parts of heat, one part from the helium side and the other part from the returning process gas in the centric tube. The heat is used to drive the chemical reaction and promote the temperature of the process gas. At the bottom of the catalyst bed, the temperature of the process gas reaches its maximum, and then, it returns to the centric tube and flows upwards. As it flows further, the heat is transferred back to the reacting process gas due to the temperature difference. Then, it leaves the reformer from the process gas outlet at the top. On the other hand, the helium with the high temperature enters the helium gas inlet at the bottom and flows upwards through the annular space formed by the catalyst tube and the guide tube. As it flows further, the heat is transferred to the reacting process gas due to the large temperature difference. Then, it leaves the reformer from the helium gas outlet at the top. Usually, in order to enhance the heat transfer, some measures are put into practice, for example, fins at the outside of the catalyst tube. Some design parameters of the reformer are listed in **Table 1**.

III. Mathematical Model of the Reformer

1. Reaction Rate Equations

Prior to High Temperature Engineering Test Reactor hydrogen production system, a catalyst test was carried out by Ohashi and Inagaki at JAERI.[11] The apparent activation energy and reaction rate constant of the Ni-based catalyst was obtained. Reaction rate equations are as follows:

Table 1 Main parameters of the steam reformer for HTR-10[10)]

Power of the reformer	2.5 MW
Hydrogen yield	2,500 Nm^3/h
Working pressure of the helium	3.0 MPa
Helium flow rate	1.6 kg/s
Outlet/inlet helium temperature	600/890°C
S (steam-to-carbon ratio)	4
Process gas flow rate	1.0 kg/s
Outlet/inlet process gas pressure at the catalyst bed	3.2/3.7 MPa
Outlet/inlet process gas temperature at the catalyst bed	840/500°C

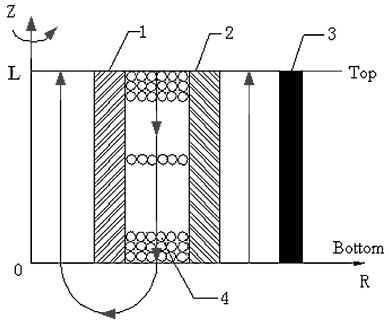


Fig. 3 Schematic diagram of the mathematic model of a single reformer tube: (1) centric tube, (2) catalyst tube, (3) guide tube and (4) catalyst bed

$$r_1 = k_1 \exp(-E_1/RT)(p_{CH_4} \cdot p_{H_2O} - p_{CO} \cdot p_{H_2}^3/K_{P_1}) \quad (4)$$

$$r_2 = k_2 \exp(-E_2/RT)(p_{CO} \cdot p_{H_2O} - p_{CO_2} \cdot p_{H_2}/K_{P_2}), \quad (5)$$

where p_{H_2}, p_{H_2O}, p_{CH_4}, p_{CO}, and p_{CO_2} are the partial pressures of H_2, H_2O, CH_4, CO, and CO_2, respectively. T and R are temperature and gas constant, respectively. The activation energy E_1 and E_2 and the pre-exponential factors k_1 and k_2 are cited from Ohashi and Inagaki.[11)] Chemical reaction constant K_{p1} and K_{p2} are cited from Hou.[12)]

2. Reformer Model

The complicated chemical reactions occur in the reformer with the process of mass, heat, and momentum transferring. **Figure 3** is the schematic diagram of the mathematic model of a single reformer tube. Since the geometry of the reformer tube is symmetrical, only right side of the tube is presented in Fig. 3. For the purpose of the model development, it is assumed that:

(1) steady-state conditions are present;
(2) pseudo-homogeneous one-dimensional model is assumed in the catalyst bed;
(3) a single reformer tube is representative of any other tube in the reformer;
(4) a heat conduction in the axial direction is negligible;
(5) a guide tube is assumed to be isothermal.

3. Mass Equation

The mass balance equations are:

$$\text{Returning process gas: } \frac{d(\rho_{1,i}u_1)}{\partial z} = 0. \quad (6)$$

$$\text{Reacting process gas: } \frac{d(\rho_{2,i}u_2)}{dz} + RC_i = 0. \quad (7)$$

$$\text{Helium: } \frac{d(\rho_{3,i}u_3)}{dz} = 0, \quad (8)$$

with the boundary conditions at:

$$z = 0,\ u_1 = u_2,\ u_3 = u_3^{inlet},\ \rho_{1,i} = \rho_{2,i},\ \rho_{3,i} = \rho_{3,i}^{inlet}$$

$$z = L,\ u_2 = u_2^{inlet},\ \rho_{2,i} = \rho_{2,i}^{inlet},$$

where z is axial coordinate; u, ρ, RC are the velocity, density, and reaction rate, respectively; subscripts 1, 2, and 3 denote returning process gas, reacting process gas, and helium, respectively, the subscript i represents gas species, *i.e.*, CH_4, H_2O, CO, H_2, CO_2, O_2, and N_2.

4. Energy Equation

The energy balance equations are expressed as follows:

Returning process gas:

$$U_1S_1(T_2 - T_1) + \frac{d(\rho_1 u_1 c_{p1} T_1)}{dz} = 0. \quad (9)$$

Reacting process gas:

$$U_1S_1(T_1 - T_2) + U_2S_3(T_3 - T_2) + (1-\varepsilon)\sum_{j=1}^{2} r_j \Delta H_j + \frac{d(\rho_2 u_2 c_{p2} T_2)}{dz} = 0. \quad (10)$$

$$\text{Helium: } U_2S_3(T_2 - T_3) + \frac{d(\rho_3 u_3 c_{p3} T_3)}{dz} = 0, \quad (11)$$

with the boundary conditions at:

$$z = 0,\ T_1 = T_2,\ T_3 = T_3^{inlet}$$

$$z = L,\ T_2 = T_2^{inlet},$$

where U, S, C_p, ε are overall heat transfer coefficient, heat transfer area per volume, specific heat, and void fraction of the catalyst bed, respectively. ΔH_j is the heat of reaction j. U_1 is calculated as follows:

$$\frac{1}{U_1} = \frac{1}{h_1} + \frac{b}{\lambda} + \frac{1}{h_2}, \quad (12)$$

where b is the thickness of the reformer wall, λ is the wall heat conduction coefficient. h_1 and h_2 are the heat transfer coefficients on the inside and outside of the centric tube wall, respectively. h_1 is calculated by Dittus-Boelter correlation. h_2 is calculated by using the correlation reported in Ref. 13). U_2 is calculated as follows:

$$\frac{1}{U_2} = \frac{1}{h_2} + \frac{b}{\lambda} + \frac{1}{h_3}, \quad (13)$$

where h_3 is the heat transfer coefficient on the outside of the catalyst tube wall. h_3 is calculated by using the Knudsen correlation[14)] if the finned catalyst tube is used.

5. Momentum Equation

The frictional pressure gradient is considered here while the accelerational pressure gradient and the pressure gradient due to gravity are neglected. The pressure gradient of the

product gas and helium is calculated according to the Blasius equation:

$$\frac{\mathrm{d}p}{\mathrm{d}z} = \frac{0.3164}{\mathrm{Re}} \cdot \frac{\rho u^2}{2d}, \tag{14}$$

where p and d are the pressure and the inner diameter of the tube, respectively. The pressure gradient along the catalyst bed is calculated according to the Ergun equation.[15)]

$$\frac{\mathrm{d}p_2}{\mathrm{d}z} = \left(\frac{150}{Re} + 1.75\right) \frac{u_2^2 \rho_2}{d_2} \cdot \frac{(1-\varepsilon)}{\varepsilon^3}, \tag{15}$$

with the boundary conditions at:

$$z = 0,\ p_1 = p_2,\ p_3 = p_3^{\mathrm{inlet}}$$
$$z = L,\ p_2 = p_2^{\mathrm{inlet}},$$

where d_2 is the inner diameter of the catalyst tube. The sets of the above ordinary differential equations are discretized by means of implicit finite differences. The resulting non-linear system of algebraic equations is solved through the Newton iteration algorithm. A computer program is developed based on this model.

IV. Validation of Model

Before validating the model, the following functions are defined:

Methane conversion:

$$X_1 = \frac{F_{\mathrm{CH_4}}^{\mathrm{inlet}} - F_{\mathrm{CH_4}}^{\mathrm{outlet}}}{F_{\mathrm{CH_4}}^{\mathrm{inlet}}} = \frac{p_{\mathrm{CO}}^{\mathrm{outlet}} + p_{\mathrm{CO_2}}^{\mathrm{outlet}}}{p_{\mathrm{CO}}^{\mathrm{outlet}} + p_{\mathrm{CO_2}}^{\mathrm{outlet}} + p_{\mathrm{CH_4}}^{\mathrm{outlet}}} \times 100\%. \tag{16}$$

Carbon monoxide conversion:

$$X_2 = \frac{p_{\mathrm{CO_2}}^{\mathrm{outlet}}}{p_{\mathrm{CO}}^{\mathrm{outlet}} + p_{\mathrm{CO_2}}^{\mathrm{outlet}}} \times 100\%. \tag{17}$$

Hydrogen yield: $Y_{\mathrm{H_2}} = F_{\mathrm{H_2}}^{\mathrm{outlet}}$,

where F is the mole flow rate. Among the above functions, methane conversion and hydrogen yield are two of the most important indexes of the performance. Hydrogen yield is mainly determined by methane conversion as well as the methane flow rate. The performance test of the reformer with a single full-scale reformer tube was conducted by JAERI between the years 2001 and 2002. The catalyst zone of the reformer tube is about 6.0 m in length.[16)] In order to verify the mathematical model, simulation results have been compared with the experimental results of JAEA. The simulation results and experimental data are listed in **Table 2**. The simulation results are shown to be in reasonable agreement with those measured in the experiment.

Another validation is the temperature distribution in the reformer tube plotted in **Fig. 4**, where good agreement between the simulating and experimental data is obtained. Figure 4 shows a drop in the temperature of about 3°C near the entrance of the catalyst tube. The reaction rate is very large at the beginning of the catalyst region because of the high methane content in the process gas, so, a large quantity of the heat is needed. Because the heat from the helium and returning process gas is not enough to drive the reaction, the process gas reduces its temperature to bring an extra heat. As the process gas flows further, the reaction rate decreases with the decreasing of the methane content and the decreasing of temperature of the process gas. When the heat from the helium and returning process gas equals to the heat for driving the reaction, the temperature of the process gas reaches its minimum. Thereafter, the heat from the helium and returning process gas is in surplus to drive the reaction and the extra heat is used to heat the process gas, so, the temperature increases. Although the temperature increases, the reaction rate decreases. This is because the methane content is more dominant than the temperature. It reaches the maximum temperature, namely, the ultimate reforming temperature at the bottom, where maximum methane conversion is obtained. The ultimate reforming temperature indicates the equilibrium limit of the methane conversion. The higher the tem-

Table 2 Comparison of the simulation results with the experiment data

	Experiment	Simulation
Inlet parameter:		
$F_{\mathrm{CH_4}}$ (mol/h)	2,700	2,700
$\mathrm{H_2O/CH_4}$	3.45	3.45
T_2^{inlet} (°C)	520	520
T_3^{inlet} (°C)	851.2	851.2
P_2^{inlet} (MPa)	4.26	4.26
P_3^{inlet} (MPa)	4.14	4.14
Outlet parameter:		
$X_{\mathrm{CH_4}}$ (%)	54.8	52.0
X_{CO} (%)	62.5	63.5
$Y_{\mathrm{H_2}}$ ($\mathrm{Nm^3/h}$)	120.2	116.4
T_3^{outlet} (°C)	638.0	651.8
T_1^{outlet} (°C)	663.0	684.1
Pressure loss:		
$(p_3^{\mathrm{inlet}} - p_3^{\mathrm{outlet}})$ (MPa)	0.056	0.053
$(p_2^{\mathrm{inlet}} - p_1^{\mathrm{outlet}})$ (MPa)	0.121	0.118

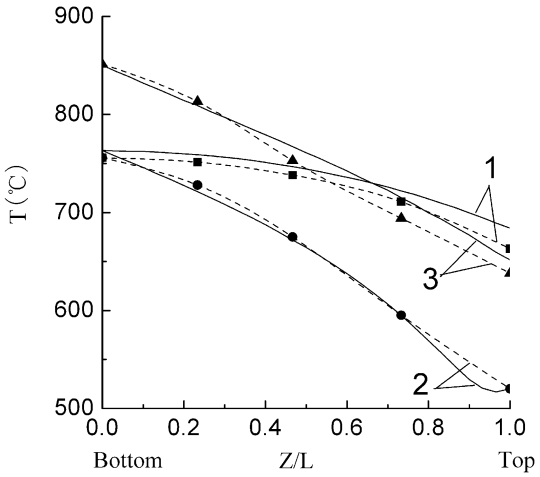


Fig. 4 Temperature distribution in the reformer tube: (dash lines) measured, (solid lines) simulated, (1) returning process gas in the centric tube, (2) reacting process gas in catalyst tube and (3) helium

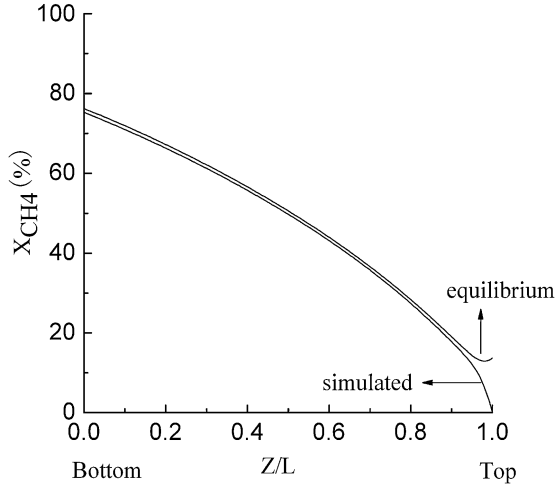


Fig. 5 Methane conversion distribution in the reformer tube

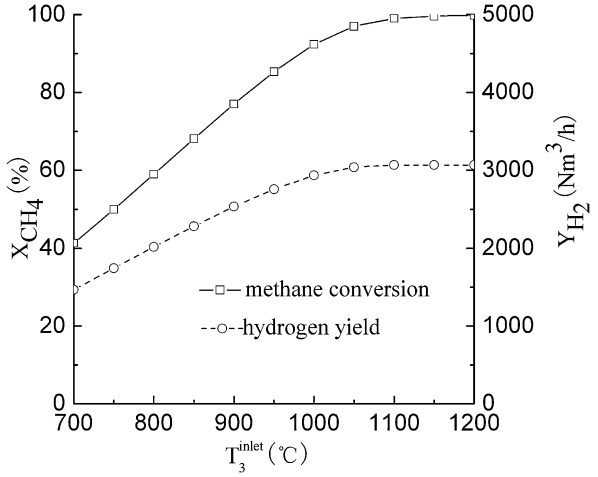


Fig. 6 Performance of the reformer with different inlet helium temperature

perature, the higher the equilibrium limit can reach. The reaction process ends at the bottom of the catalyst tube, then, the process gas turns upward to the centric tube, where the heat is transferred back to the reacting process gas with the temperature decreasing smoothly.

V. Results and Discussions

1. Methane Conversion Distribution

Using parameters listed in Table 1, the methane conversion distribution in the reformer tube is calculated. In **Fig. 5**, the equilibrium methane conversion is calculated by assuming that the thermodynamic equilibrium is reached in the catalyst under the same conditions. It provides a limit value for the methane conversion, which is independent of the catalyst performance and reaction rate. Figure 5 indicates that there exists a large difference between the equilibrium and simulating data at the beginning of the catalyst region because the residence time is not long enough. As the process gas flows further, the residence time increases, then, the conversion increases rapidly toward an equilibrium value due to the large reaction rate. The conversion reaches 78% at the bottom. With the design parameters presented in Table 1, the reformer can obtain good performance. It is reasonable to choose HTR-10 as a heat resource of the hydrogen production system.

2. The Effect of the Inlet Helium Parameters

The influence of the inlet helium temperature on the performance of the reformer is illustrated in **Fig. 6**. It shows that, at the temperature lower than 1,000°C, the methane conversion and hydrogen yield increase remarkably with the increase in the inlet helium temperature due to the endothermic nature of the reaction system. Hence, the inlet helium temperature plays an important role in the reformer performance. As the inlet helium temperature increases, the ultimate reforming temperature increases, resulting in the improvement of the methane conversion and hydrogen yield. So, in order to enhance the performance of the reformer, an important way is to increase the inlet helium temperature,

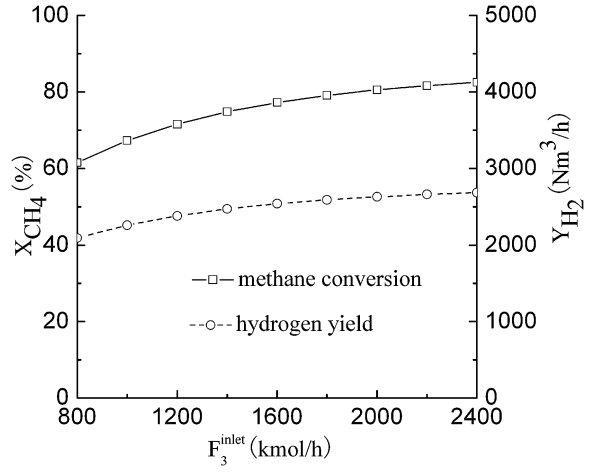


Fig. 7 Performance of the reformer with different inlet helium flow rate

i.e., the outlet gas temperature of the reactor while it is limited by the core temperature due to the limitation of the core material. In addition, the heat loss occurring in the pipes from the outlet of IHX to the inlet of the reformer should be kept at the minimum. However, when the temperature is higher than 1,000°C, the influence becomes weak because the methane conversion approaches 100%.

Figure 7 shows the effect of the inlet helium flow rate on the performance. It is obviously seen that the performance increases with the increase in the inlet helium flow rate. As the helium flow rate increases, the heat transfer coefficient on the outside of the catalyst tube wall increases, which enhances the heat transferring to the catalyst bed. On the other hand, the high helium flow rate introduces a large pressure loss.

Figure 8 shows the effect of the inlet helium pressure on the performance. It does not show any change of the performance. This is because the helium flow rate and the helium physical properties such as heat conduction coefficient, heat capacity, and viscosity coefficient almost remain con-

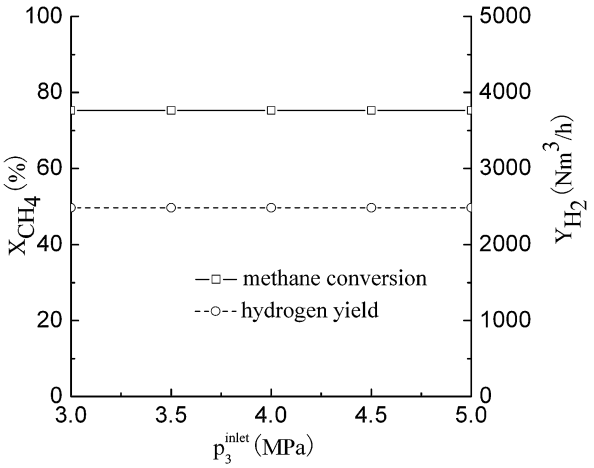


Fig. 8 Performance of the reformer with different inlet helium pressure

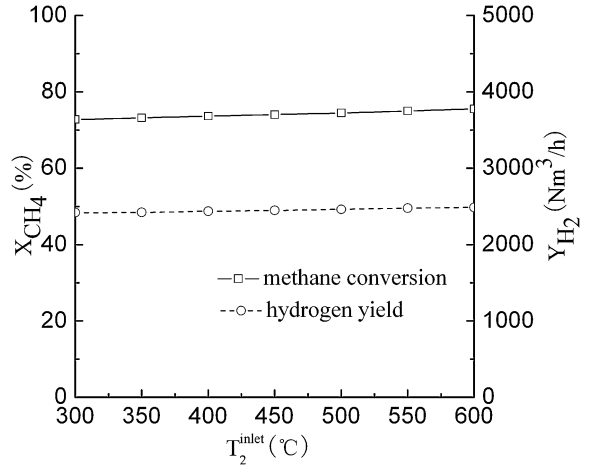


Fig. 9 Performance of the reformer with different inlet process gas temperature

stant within the pressure range investigated. Consequently, the heat transfer coefficient on the outside of the catalyst tube wall almost remains constant. The ultimate result is that the heat transferred to the process gas remains unchanged.

3. The Effect of the Inlet Process Gas Parameters

Figure 9 shows the temperature of the inlet process gas has almost no influence on the performance. This is because the ultimate reforming temperature keeps almost unchanged, which can be seen from **Fig. 10**. The temperature of the inlet process gas increases from 300°C to 600°C while the ultimate reforming temperature increases only about 11°C. It demonstrates that the increase of the inlet process gas temperature does not lead to a pronounced improvement of the performance.

The effect of the flow rate of the inlet process gas on the reformer performance is illustrated in **Fig. 11**. It indicates that the methane conversion decreases with the increase in the process gas flow rate since the ultimate reforming temperature is reduced, which can be seen from **Fig. 12**. However, within the range investigated, the hydrogen yield in-

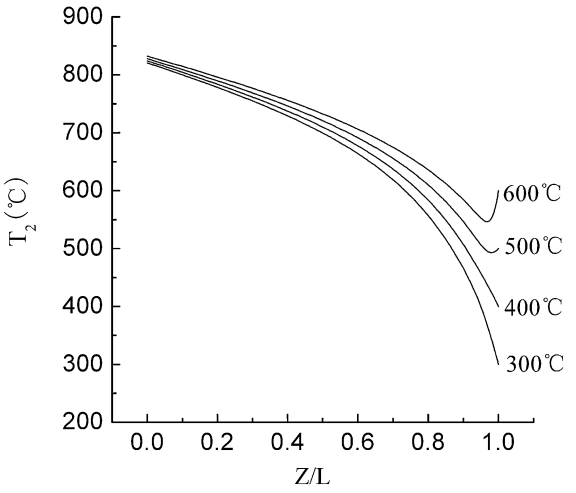


Fig. 10 Temperature distribution in the catalyst tube with different inlet process gas temperature

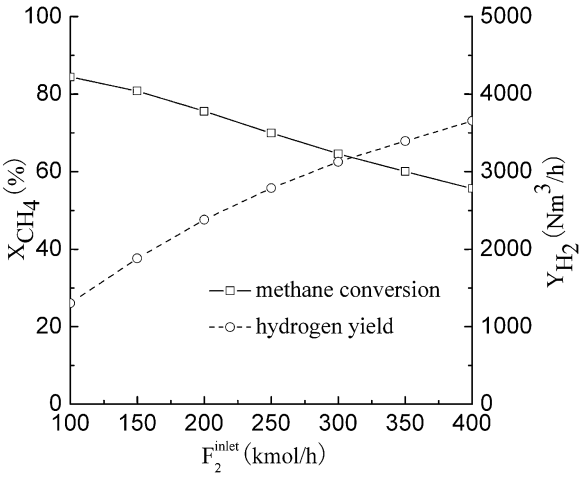


Fig. 11 Performance of the reformer with different inlet process gas flow rate

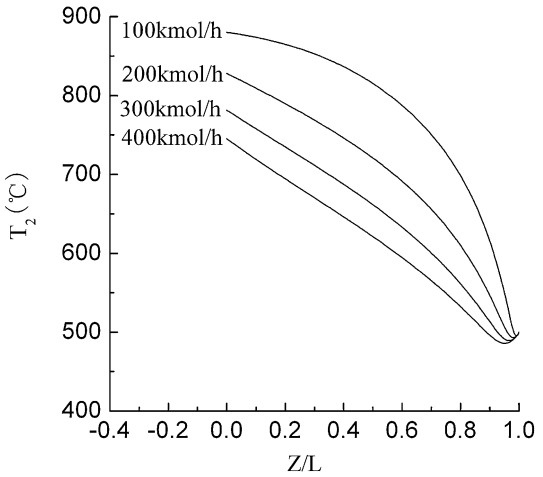


Fig. 12 Temperature distribution in the catalyst tube with different inlet process gas flow rate

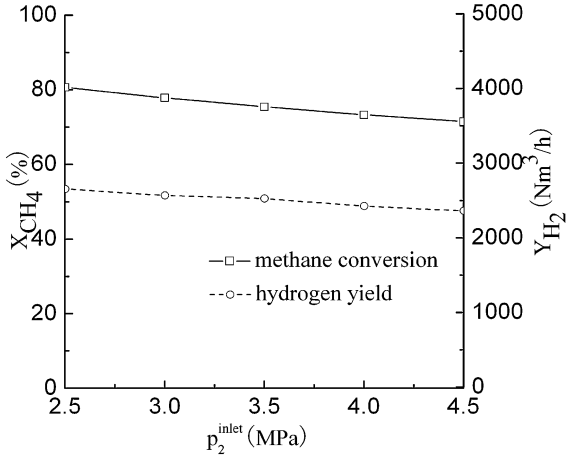


Fig. 13 Performance of the reformer with different inlet process gas pressure

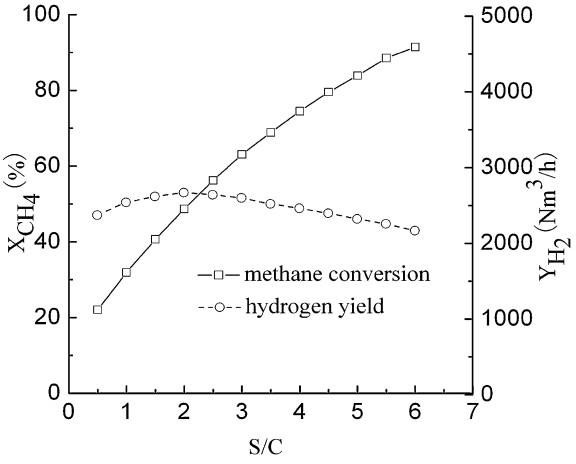


Fig. 14 Performance of the reformer with different S/C

creases with the increase in the inlet process gas flow rate. This is because the heat transfer coefficient of the catalyst tube wall increases with the process gas flow rate, which improves the heat transferring to the process gas. But, the high flow rate lowers the utilization rate of the methane. So, an optimal flow rate should be chosen.

Figure 13 illustrates that the methane conversion and hydrogen yield decrease with the increase of the inlet process gas pressure due to an increase the number of moles of the reaction system. Considering the permeation of the radioactive materials, the pressure of the process gas has to be higher than that of the secondary helium loop, which also has to be higher than that of the primary helium loop. However, the pressure of the primary helium loop is determined by the nuclear reactor system.

4. The Effect of S/C

The effect of S/C on the performance is demonstrated in **Fig. 14**. The methane conversion increases with the increase in S/C due to the enhancement of the steam content in the process gas, which shifts reaction (1) and (2) toward the right. When S/C increases, the hydrogen yield increases to its maximum at the neighborhood of (2), but, then, decreases with the increase in S/C. This is because the hydrogen yield is determined by the methane conversion as well as the methane content. When S/C is at a lower value, the methane conversion is low. In addition, a lower value of S/C leads carbon formation in the catalyst bed. On the other hand, when S/C is at a higher value, the methane content is low. Additionally, higher S/C means a higher steam content. We must supply the energy to make water as reactants at room temperature into steam. But we can not recover the energy of extra steam after the SMR process, which means the energy consumption will be higher with the higher value of S/C. Consequently, S/C is always kept at an optimal value, *e.g.*, 4.0.

VI. Conclusions

Based on the pseudo-homogeneous one-dimensional model, a steady-state mathematical model has been developed. Good agreement is obtained between the simulating results and experimental data.

Thermodynamic and chemical kinetic analysis shows that it is reasonable to choose HTR-10 as a heat resource of the hydrogen production system. With design parameters presented in this paper, the reformer can obtain a high performance.

The inlet helium temperature influences the performance remarkably when it is lower than 1,000°C. However, the effect becomes weak when the temperature is higher than 1,000°C. The influence of the inlet helium flow rate is weaker than that of the temperature. The inlet helium pressure and inlet process gas temperature have almost no influence on the performance. When the flow rate of the inlet process gas increases, the hydrogen yield increases while the methane conversion drops. The performance increases with the decrease in the inlet process gas pressure. So, it is suggested to design HTR-10 operating at a higher exit temperature and a lower system pressure, if possible. The influence of S/C is complicated. The methane conversion increases and the hydrogen yield decreases as S/C increases. S/C is always kept at an optimal value, *e.g.*, 4.0.

The coupling technology between the hydrogen production system and HTR-10, and the safety issues such as dynamic characteristics of the system will be investigated further.

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